

Arkadip Som

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EDUCATION

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech in Civil Engineering

Howrah, West Bengal, India

Aug 2024 – July 2028

- **CGPA:** 7.99 / 10.00 (Till 4th Semester)

Begampur High School

Higher Secondary (Class XII) & Secondary (Class X)

Hooghly, West Bengal, India

Jan 2015 – Apr 2023

- **Higher Secondary Examination (WBCHSE) - 2023:** Secured **87.8%** aggregate; ranked **2nd** institutionally.
- **Secondary Examination (WBBSE) - 2021:** Secured **97.8%** aggregate; ranked **12th across West Bengal** and **2nd** institutionally.

TECHNICAL & STRUCTURAL PROJECTS

Predictive Pavement Asset Management System | [\[Live Link\]](#) | [\[GitHub\]](#)

Python, Scikit-Learn, Streamlit, Plotly

Independent Project

Sept 2026

- Developed an end-to-end Pavement Management System utilizing **Gradient Boosting Regression** to predict long-term structural and functional road deterioration (**IRI, PCI**) based on **cumulative traffic loads (ESALs / MSA)** and **subgrade CBR properties**.
- Automated capital planning using **IRC:37** and **NHAI maintenance guidelines**, dynamically mapping predicted surface distress to multi-year budget schedules across preventive maintenance, **bituminous overlays**, and full depth reconstruction.
- Architected a full-stack analytical platform integrating empirical axle-load performance models, compound traffic projections, and dynamic **Plotly financial dashboards** to optimize highway lifecycle maintenance expenditures.

Spatial Analysis & Terrain Modeling

ArcMap (ArcGIS), ERDAS Imagine, DEM Data

Coursework Project (CE 2274) - 4th Semester

Jan 2026 - May 2026

- Conducted **georeferencing, vectorization and mosaicking** of toposheets to extract precise areas of interest and build vector/raster datasets.
- Processed **Digital Elevation Model (DEM)** datasets for **catchment delineation**, computing watershed characteristics, and estimating average rainfall for hydrological modeling.
- Executed **overlay and proximity spatial analysis** to evaluate geographical features and generate map layouts.

Reinforcement Detailing of G+1 RC Building

IS 456, IS 875, SP-16, SP-34

Coursework Project (CE 3172) - 5th Semester

July 2026 - Present

- Executing complete structural load estimations (**Dead & Live Load Combinations**) and **Limit State Design** for a G+1 RC frame structure as per **IS 875 (Parts 1 & 2)** and **IS 456:2000**.
- Designing critical structural members (**one & two-way slabs, continuous beams, columns, staircase, isolated & combined footings, piles & pile caps**) using **SP-16** and preparing **detailing drawings** and **BBS** as per **SP-34**.

INDUSTRIAL CASE STUDIES

Bureau of Indian Standards (BIS) — Industrial Case Study

UltraTech Cement Plant, Dankuni

Team Lead & Field Researcher

Jan 2026 – April 2026

- Led a **3-member team** in a national BIS case study, conducting on-site **quality compliance audits of cement manufacturing and lab testing workflows** at UltraTech Cement (Dankuni Unit).
- Evaluated plant Quality Assurance routines against Indian Standards (**IS 269, IS 1489, IS 455**) and co-authored a **100-page compliance report** identifying critical process benchmarks.

POSITION OF RESPONSIBILITY

- **Publicity Team, INSTRUO 14.0 (Techno-Management Fest)** — led outreach at Jadavpur University Salt Lake campus for 30+ events, with focused promotion of Hacksprint, the flagship hackathon (5L prize pool, 55% of the fest's 9L total)

TECHNICAL SKILLS

- **Core Coursework:** Transportation & Pavement Engineering, GIS & Spatial Analysis, Structural Analysis, RCC Design, Geotechnical Engineering, Water Resources Engineering
- **Software & GIS Tools:** ArcMap (ArcGIS), ERDAS Imagine, AutoCAD (2D Drafting), MS Office, Git/GitHub
- **Programming & Analytics:** Python, NumPy, Pandas, Matplotlib, Scikit-Learn, Streamlit, SQL
- **Languages:** English (Fluent), Bengali (Native), Hindi (Conversational)